
QUAID-E-AWAM UNIVERSITY OF ENGINEERING, SCIENCE & TECHNOLOGY

Database Management Systems

Week 2, Day 6

MySQL—A RDBMS

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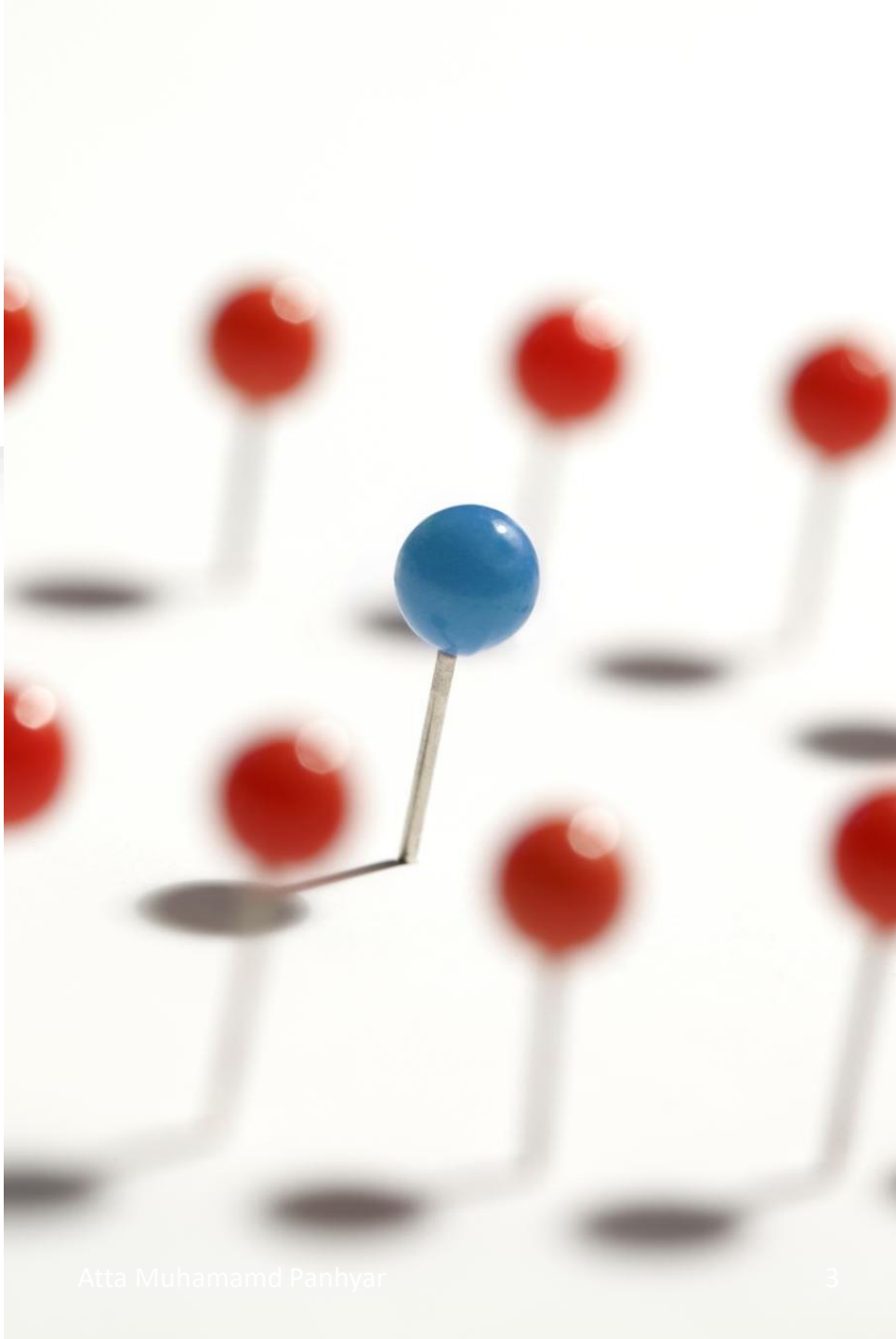
RECAP

- Having Clause
- Inner Join
- LEFT/RIGHT JOIN



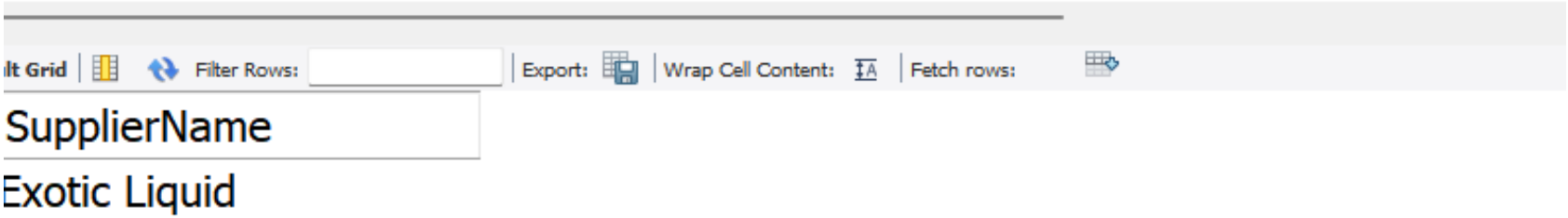
EXISTS Operator

- The EXISTS operator is used to test for the existence of any record in a subquery.

- The EXISTS operator returns TRUE if the subquery returns one or more records.

EXISTS Example

```
1 •  SELECT SupplierName FROM Suppliers
2    WHERE     EXISTS( SELECT ProductName FROM Products
3            WHERE Products.SupplierID = Suppliers.supplierID
4                    AND Price < 20);
```



CASE

- The CASE statement goes through conditions and returns a value when the first condition is met (like an if-then-else statement).
- So, once a condition is true, it will stop reading and return the result.
- If no conditions are true, it returns the value in the ELSE clause.

CASE –Example

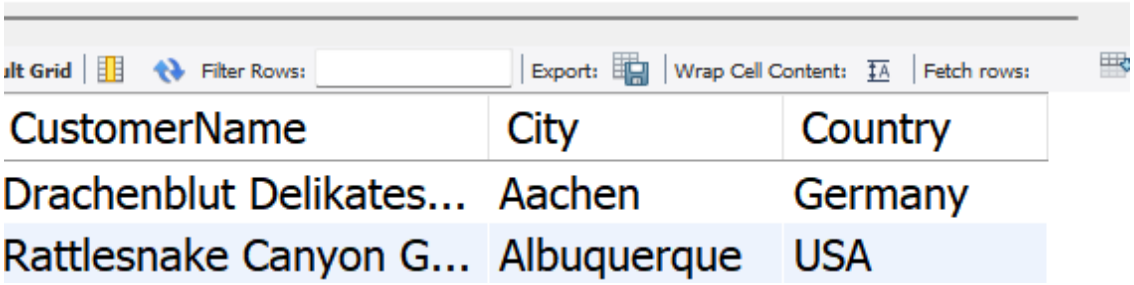
```
1 •  SELECT OrderID, Quantity,
2    CASE
3        WHEN Quantity > 30 THEN 'The quantity is greater than 30'
4        WHEN Quantity = 30 THEN 'The quantity is 30'
5        ELSE 'The quantity is under 30'
6    END AS QuantityText
7    FROM OrderDetails;
```

lt Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

OrderID	Quantity	QuantityText
10248	12	The quantity is under ...
10248	10	The quantity is under ...

CASE –Example

```
1 •  SELECT CustomerName, City, Country
2    FROM Customers
3    ORDER BY
4    (CASE
5        WHEN City IS NULL THEN Country
6        ELSE City
7    END);
```

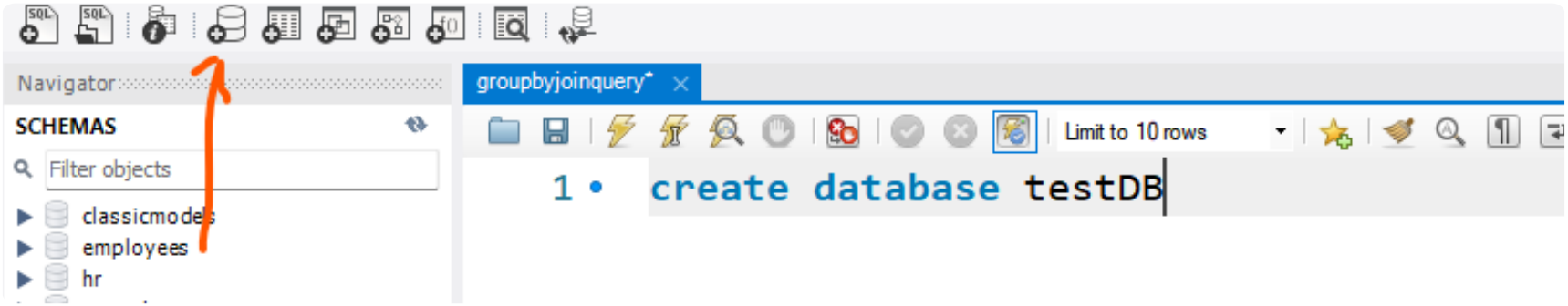


CustomerName	City	Country
Drachenblut Delikates...	Aachen	Germany
Rattlesnake Canyon G...	Albuquerque	USA

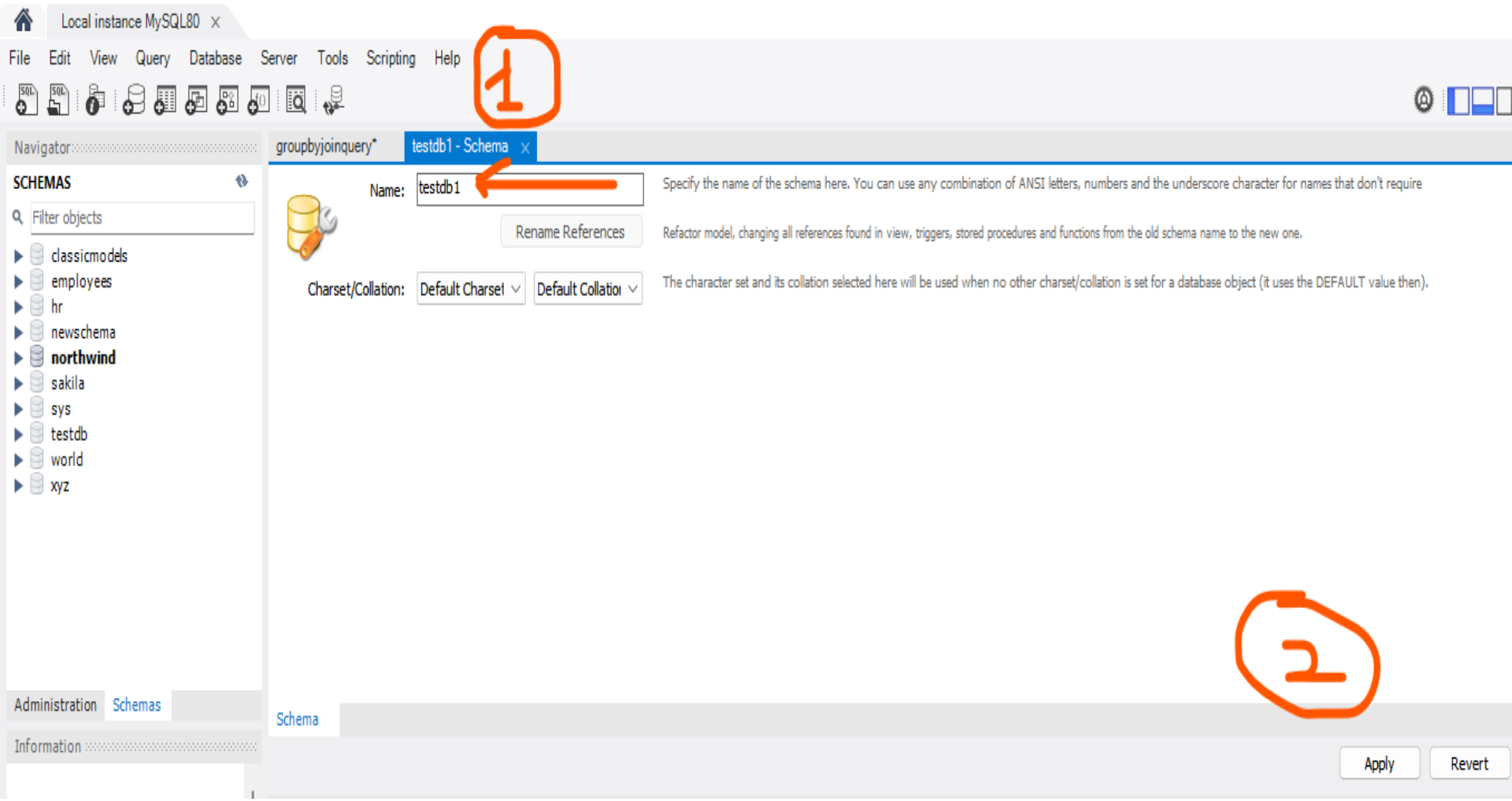
Create a Database

- The CREATE DATABASE statement is used to create a new SQL database.

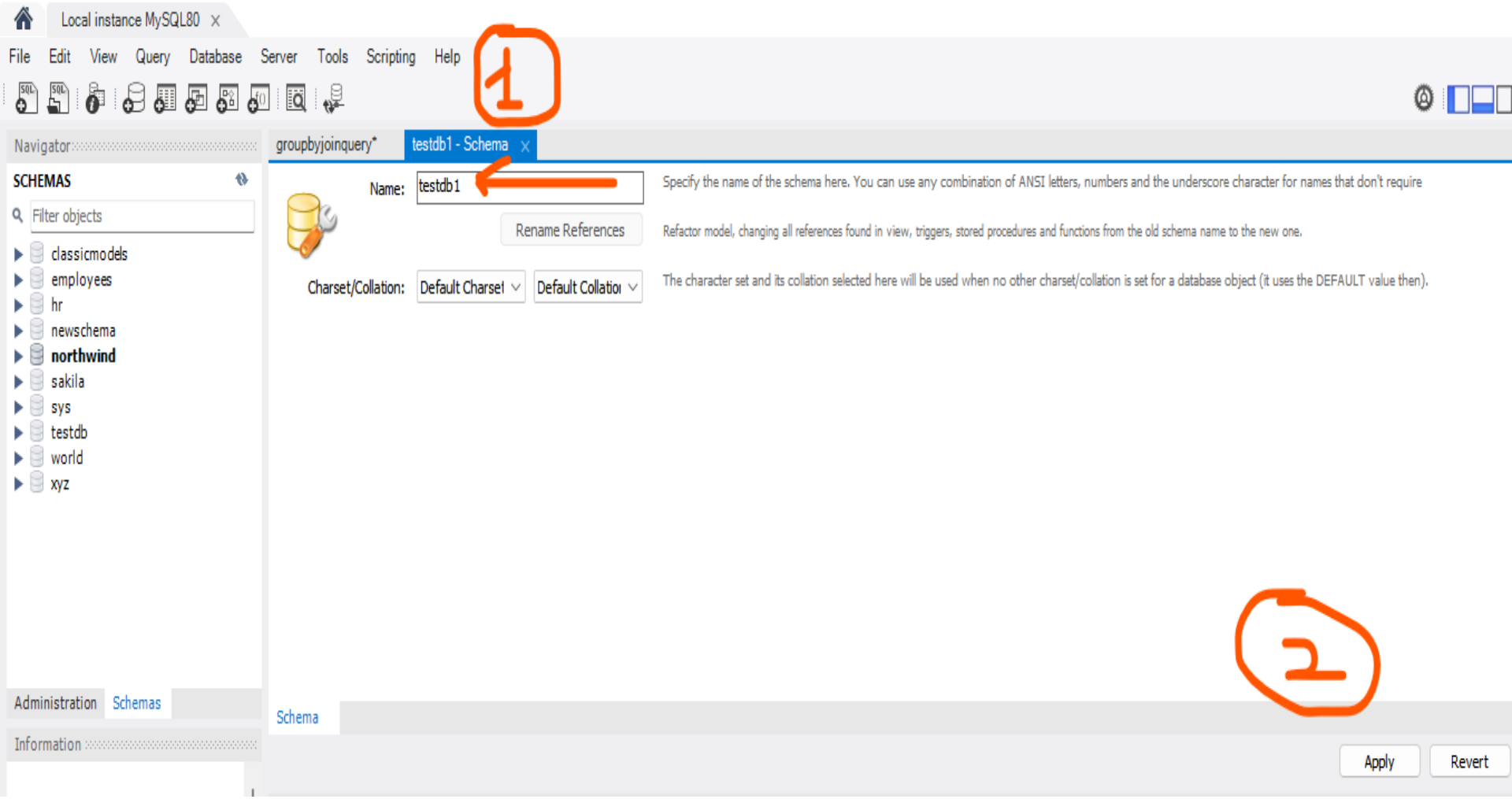
```
1 •  create database testDB
```



CREATE a DATABASE


Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
1
Navigator
SCHEMAS
Filter objects
classicmodels
employees
hr
newschema
northwind
sakila
sys
testdb
world
xyz
Administration Schemas
Information
groupbyjoinquery*
testdb1 - Schema
Name: testdb1
Specify the name of the schema here. You can use any combination of ANSI letters, numbers and the underscore character for names that don't require
Rename References
Refactor model, changing all references found in view, triggers, stored procedures and functions from the old schema name to the new one.
Charset/Collation: Default Charset Default Collatio
The character set and its collation selected here will be used when no other charset/collation is set for a database object (it uses the DEFAULT value then).
2
Schema
Apply
Revert

DATABASE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Revert

DATABASE

Apply SQL Script to Database

Review SQL Script

Apply SQL Script

Review the SQL Script to be Applied on the Database

Online DDL

Algorithm: Default

Lock Type: Default

```
1    CREATE SCHEMA `testdb1` ;
2
```

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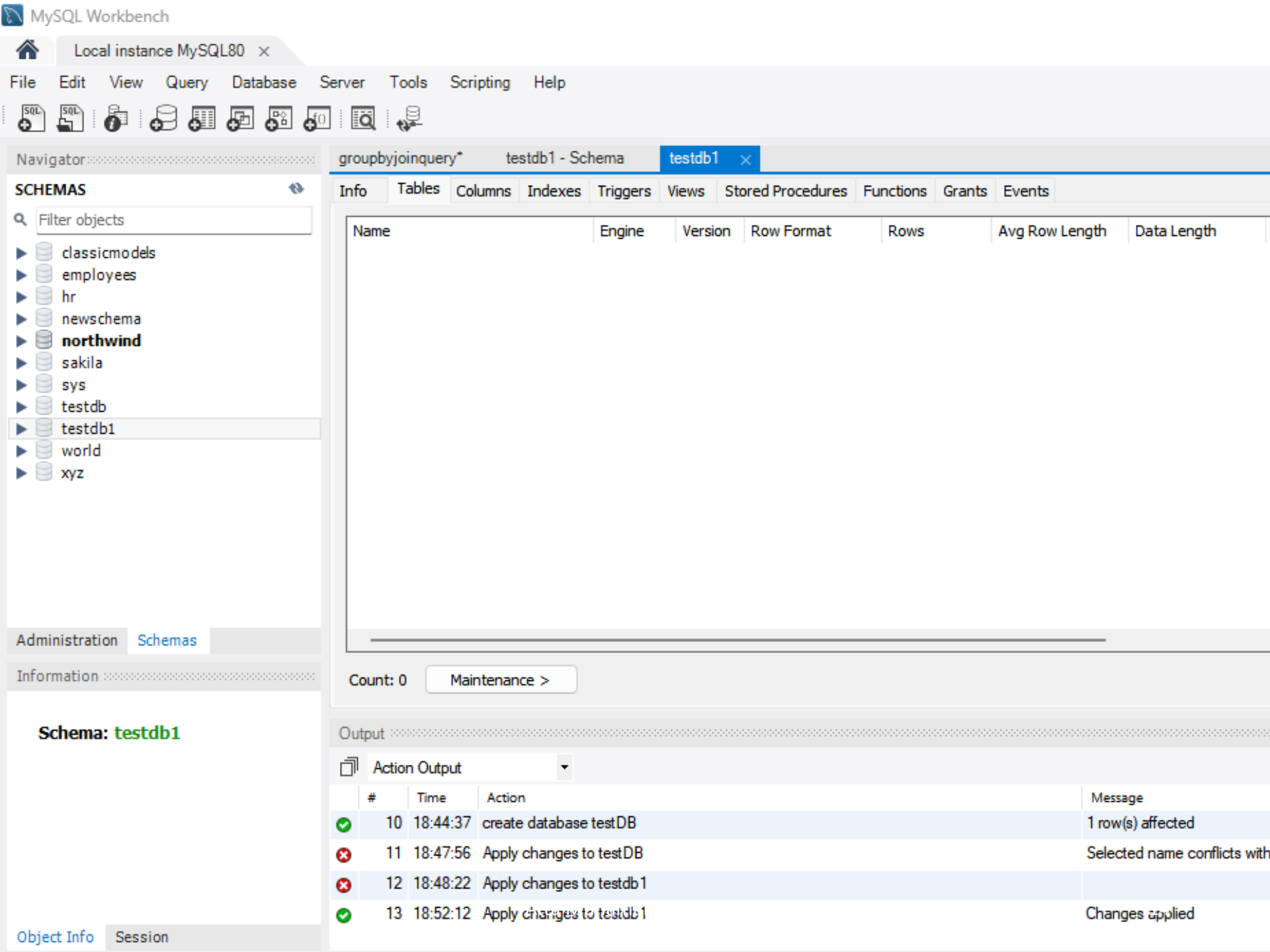
Back | Apply | Cancel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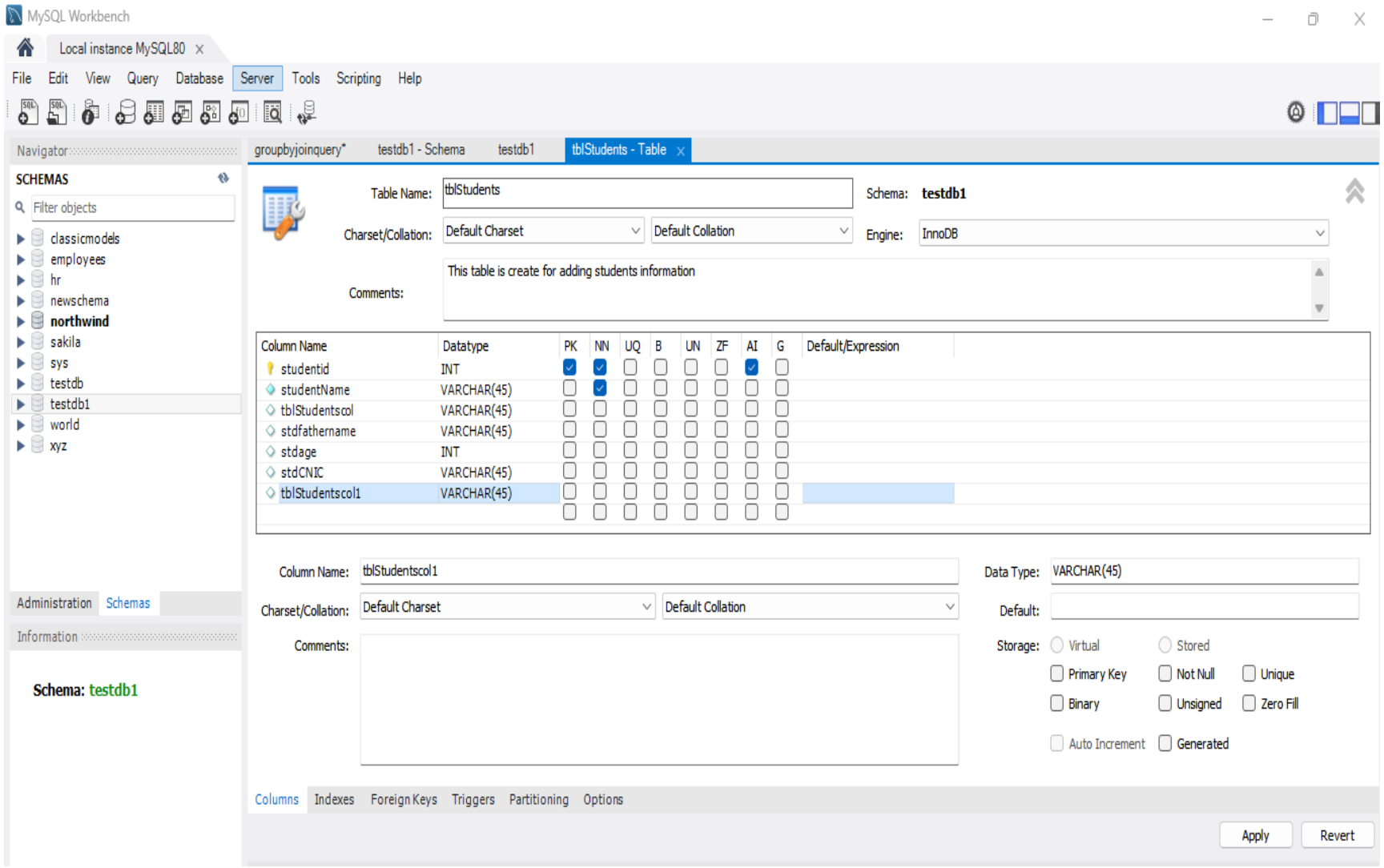
testdb
testdb1
world
xyz
Administration Schemas
Information
Schema: testdb1
Object Info Session
groupbyjoinquery*
testdb1 - Schema
testdb1
Info Tables Columns Indexes Triggers Views Stored Procedures Functions Grants Events
Name Engine Version Row Format Rows Avg Row Length Data Length
Count: 0
Maintenance >
Output
Action Output
Time Action Message
10 18:44:37 create database testDB 1 row(s) affected
11 18:47:56 Apply changes to testDB Selected name conflicts with
12 18:48:22 Apply changes to testdb1
13 18:52:12 Apply changes to testdb1 Changes applied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sys
testdb
testdb1
world
xyz
Administration Schemas
Information
Schema: testdb1
groupbyjoinquery*
testdb1 - Schema
testdb1
tblStudents - Table
Table Name: tblStudents
Schema: testdb1
Charset/Collation: Default Charset
Default Collation
Engine: InnoDB
Comments: This table is create for adding students information
Column Name
Datatype
PK
NN
UQ
B
UN
ZF
AI
G
Default/Expression
studentid INT
studentName VARCHAR(45)
tblStudentscol VARCHAR(45)
stdfathername VARCHAR(45)
stdage INT
stdCNIC VARCHAR(45)
tblStudentscol1 VARCHAR(45)
Column Name: tblStudentscol1
Data Type: VARCHAR(45)
Charset/Collation: Default Charset
Default Collation
Default:
Comments:
Storage: Virtual Stored
Primary Key Not Null Unique
Binary Unsigned Zero Fill
Auto Increment Generated
Columns Indexes Foreign Keys Triggers Partitioning Options
Apply
Revert

Apply SQL Script to Database

Review SQL Script

Apply SQL Script

Review the SQL Script to be Applied on the Database

Online DDL

Algorithm: Default Lock Type: Default

```
1  CREATE TABLE `testdb1`.`tblstudents` (
2    `studentid` INT NOT NULL AUTO_INCREMENT,
3    `studentName` VARCHAR(45) NOT NULL,
4    `tblStudentscol` VARCHAR(45) NULL,
5    `stdfathername` VARCHAR(45) NULL,
6    `stdage` INT NULL,
7    `stdCNIC` VARCHAR(45) NULL,
8    `tblStudentscol1` VARCHAR(45) NULL,
9    PRIMARY KEY (`studentid`))
10 COMMENT = 'This table is create for adding students information';
11
```

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Back Apply Cancel

Apply SQL Script to Database

Review SQL Script

Apply SQL Script

Applying SQL script to the database

The following tasks will now be executed. Please monitor the execution.
Press Show Logs to see the execution logs.

- [x] Execute SQL Statements

SQL script was successfully applied to the database.

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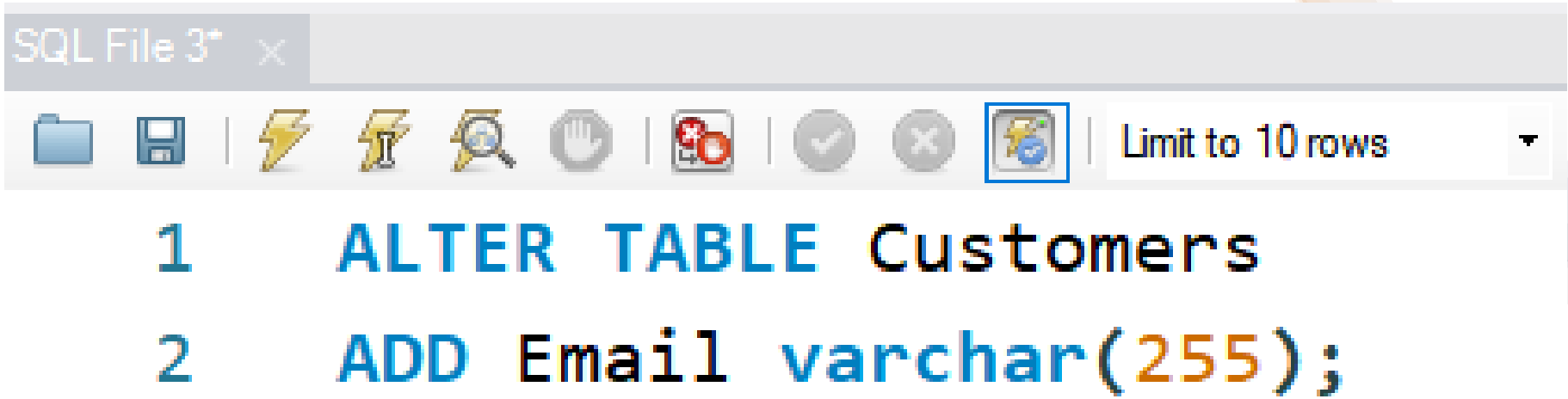
Show Logs | Back | Finish | Cancel

DROP DATABASE Statement

ALTER TABLE Statement

- The ALTER TABLE statement is used to add, delete, or modify columns in an existing table.

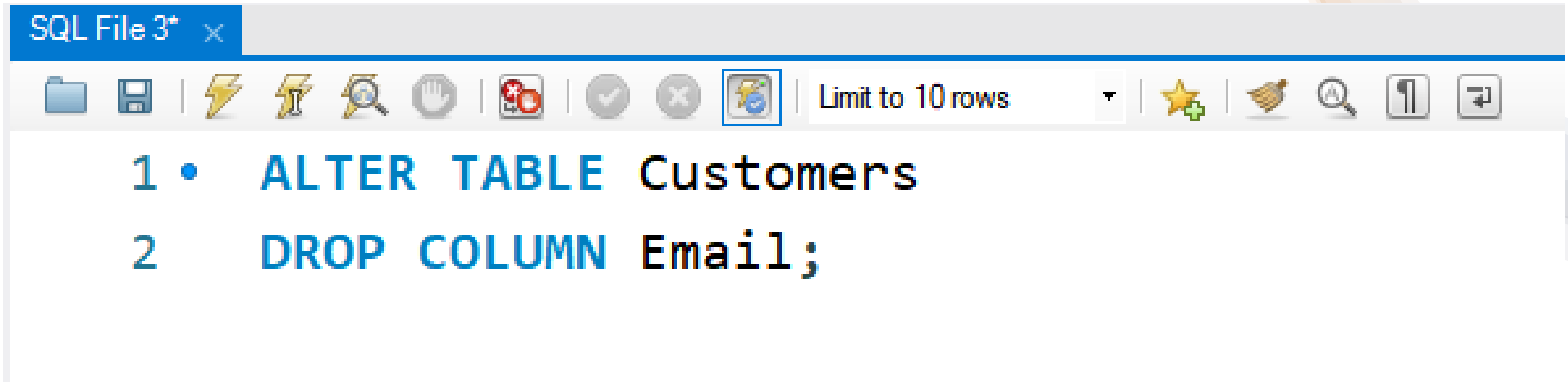
- The ALTER TABLE statement is also used to add and drop various constraints on an existing table



ALTER TABLE –Example

ALTER TABLE - DROP COLUMN

- To delete a column in a table, use the following syntax (notice that some database systems don't allow deleting a column):



ALTER TABLE –DROP COLUMN

ALTER TABLE - MODIFY COLUMN

- To change the data type of a column in a table, use the following syntax

```
1 •  ALTER TABLE customers
2    ADD DateOfBirth date;
3    -- Run these commands separetly
4 •  ALTER TABLE customers
5    MODIFY COLUMN DateOfBirth year;
```

MODIFY

Constraints

- SQL constraints are used to specify rules for data in a table.
- Constraints can be specified when the table is created with the CREATE TABLE statement, or after the table is created with the ALTER TABLE statement.

Constraints

- Constraints are used to limit the type of data that can go into a table.
- This ensures the accuracy and reliability of the data in the table.
- If there is any violation between the constraint and the data action, the action is aborted.

Constraints

- Constraints can be column level or table level.
- Column level constraints apply to a column, and table level constraints apply to the whole table.

Common Constraints

- NOT NULL - Ensures that a column cannot have a NULL value
- UNIQUE - Ensures that all values in a column are different
- PRIMARY KEY - A combination of a NOT NULL and UNIQUE. Uniquely identifies each row in a table
- FOREIGN KEY - Prevents actions that would destroy links between tables
- CHECK - Ensures that the values in a column satisfies a specific condition
- DEFAULT - Sets a default value for a column if no value is specified
- CREATE INDEX - Used to create and retrieve data from the database very quickly

NOT NULL Constraint

- By default, a column can hold NULL values.
- The NOT NULL constraint enforces a column to NOT accept NULL values.

```
CREATE TABLE Persons (
    ID int NOT NULL,
    LastName varchar(255) NOT NULL,
    FirstName varchar(255) NOT NULL,
    Age int
);
```

Constraints on ALTER TABLE

```
ALTER TABLE Persons
MODIFY Age int NOT NULL;
```

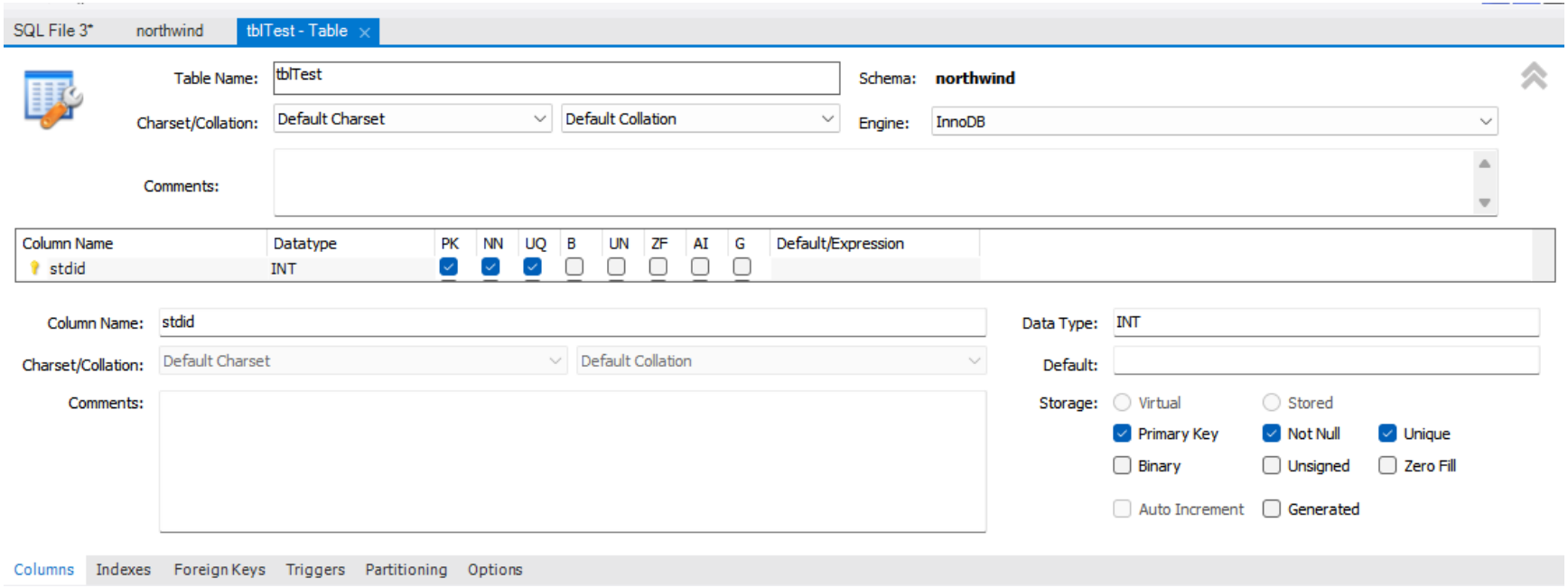
UNIQUE Constraint

- The UNIQUE constraint ensures that all values in a column are different.
- A PRIMARY KEY constraint automatically has a UNIQUE constraint.

UNIQUE Constraint on CREATE TABLE

```
CREATE TABLE Persons (
    ID int NOT NULL,
    LastName varchar(255) NOT NULL,
    FirstName varchar(255),
    Age int,
    UNIQUE (ID)
);
```

Graphically…



PRIMARY KEY Constraint

- The PRIMARY KEY constraint uniquely identifies each record in a table.
- Primary keys must contain UNIQUE values and cannot contain NULL values.
- A table can have only ONE primary key; and in the table, this primary key can consist of single or multiple columns (fields)

FOREIGN KEY Constraint

- The FOREIGN KEY constraint is used to prevent actions that would destroy links between tables.
- A FOREIGN KEY is a field (or collection of fields) in one table, that refers to the PRIMARY KEY in another table.

FOREIGN KEY

```
CREATE TABLE Orders (
    OrderID int NOT NULL,
    OrderNumber int NOT NULL,
    PersonID int,
    PRIMARY KEY (OrderID),
    FOREIGN KEY (PersonID) REFERENCES Persons(PersonID)
);
```

Question(s)